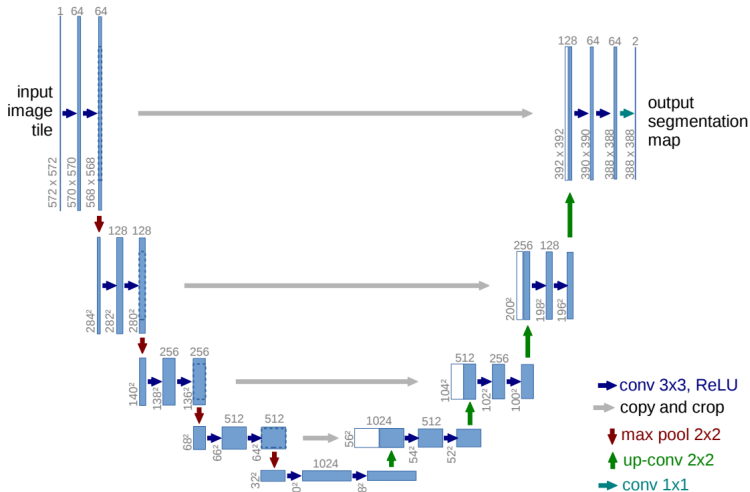


# Deep Learning

Vazgen Mikayelyan



# UNet



## 1 Kullback-Leibler Divergence

## 2 Autoencoders

# KL Divergence

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but there is no symmetry, i.e.  $KL(P||Q) \neq KL(Q||P)$ .

# Jensen-Shannon Divergence

JS Divergence is the following

$$JS(P||Q) = \frac{1}{2}KL(P||M) + \frac{1}{2}KL(M||Q),$$

where  $M = \frac{P + Q}{2}$ .

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Suppose that we have two multivariate normal distributions:

$\mathcal{N}_1(\mu_1, \Sigma_1), \mathcal{N}_2(\mu_2, \Sigma_2)$ . Then

$$KL(\mathcal{N}_1, \mathcal{N}_2) = \frac{1}{2} \left( \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) - k + \ln \frac{|\Sigma_2|}{|\Sigma_1|} \right).$$

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In one dimensional case we will have

$$KL(\mathcal{N}_1, \mathcal{N}_2) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

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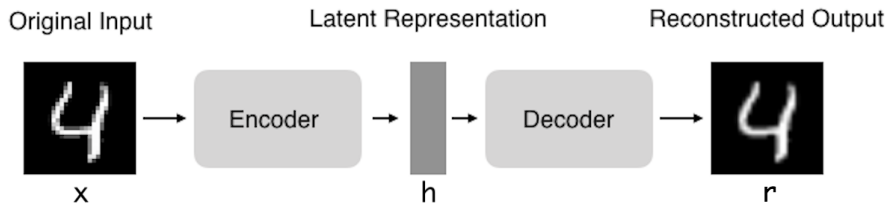
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This is the part of the network that compresses the input into a latent-space representation.

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This part aims to reconstruct the input from the latent space representation.

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